1.5 Chemical Energetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Hard

Time allowed: 20

Score: /16

Percentage: /100

Question 1a

Urea, $CO(NH_2)_2$, is a naturally occurring substance which can be hydrolysed with water to form ammonia and carbon dioxide.

The standard enthalpy changes of formation of water, urea, carbon dioxide and ammonia (in aqueous solution) are given below in Table 1.1

Table 1.1

compound	**$\Delta H^{\theta}f$ / kJ mol^{-1}**
H_2O (l)	–287.0
$CO(NH_2)_2$ (aq)	–320.5
CO_2 (g)	–414.5
NH_3 (aq)	–81.0

i)

Write an equation for the reaction.

[1]

ii)

Construct a simple energy cycle for the hydrolysis of urea.

[2]

[3 marks]

Question 1b

Use these data to calculate the standard enthalpy change for the hydrolysis of urea.

[1 mark]

Question 1c

Calculate the enthalpy of combustion of urea, given the following equation and the data in Table 1.1.

$$2CO(NH_2)_2\ (s) + 3O_2\ (g) \rightarrow 2CO_2\ (g) + 2N_2\ (g) + 4H_2O\ (l)$$

[1 mark]

Question 2a

The enthalpy change of solution for ammonium chloride can be measured using calorimetry. 12.04 g of ammonium chloride is dissolved in 125.0 cm^3 of water at 19.5 °C.

The enthalpy of solution of ammonium chloride is +15.1 kJ mol^{-1}. Determine the energy change, in J, when 12.04 g of ammonium chloride is dissolved in 125.0 cm^3 of water.

[2 marks]

Question 2b

Use your answer to part (a) to determine the change in temperature, in °C, when the ammonium chloride is dissolved.

[1 mark]

Question 2c

Use your answer to part (b) to determine the final temperature, in °C, of the solution during the reaction.

[1 mark]

Question 3a

Determine the enthalpy of hydrogenation of propene using the data in Table 3.1.

Table 3.1

Bond	Bond enthalpy / kJ mol^{-1}
H-H	435
C-H	413
C-C	347
C=C	619

Enthalpy of hydrogenation, ΔH_r = ... kJ mol^{-1}

[3 marks]

Question 3b

Use the data in Table 3.1 to suggest, with a reason, whether the polymerisation of propene is exothermic or endothermic

[2 marks]

Question 3c

Carbon, hydrogen and ethene each burn exothermically in an excess of air.

$C\ (s) + O_2\ (g) \rightarrow CO_2\ (g)$	$\Delta H^\theta{}_c = -393.7\ \text{kJ mol}^{-1}$
$H_2\ (g) + \frac{1}{2}O_2\ (g) \rightarrow H_2O\ (l)$	$\Delta H^\theta{}_c = -285.9\ \text{kJ mol}^{-1}$
$C_2H_4\ (g) + 3O_2\ (g) \rightarrow 2CO_2\ (g) + 2H_2O\ (l)$	$\Delta H^\theta{}_c = -1411.0\ \text{kJ mol}^{-1}$

Use the data to calculate the standard enthalpy change of formation, $\Delta H^\theta{}_f$, in kJ mol^{-1}, of ethene at 298 K.

$$2C\ (s) + 2H_2\ (g) \rightarrow C_2H_4\ (g)$$

$\Delta H^\theta{}_f$ = ... kJ mol^{-1}

[2 marks]

Model Answers: Hard

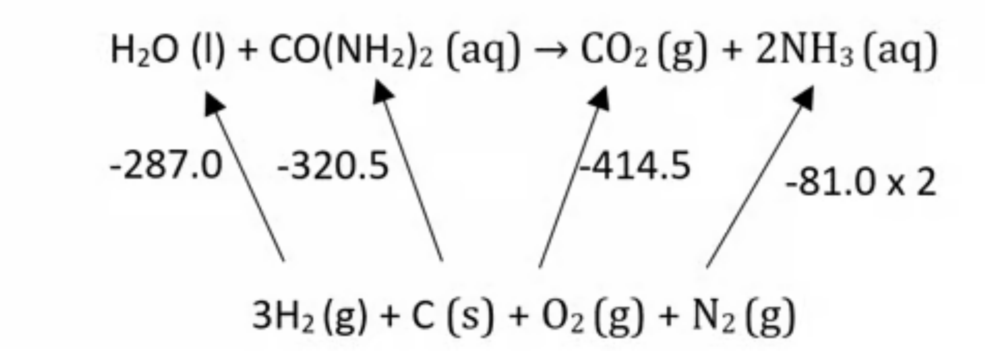
1a

a)

i) The equation for the hydrolysis of urea by water is:

- H_2O (l) + $CO(NH_2)_2$ (aq) → CO_2 (g) + $2NH_3$ (aq); [1 mark]

ii) A simple energy cycle for the hydrolysis of urea is:



- Arrows pointing upwards; [1 mark]
- Correct substances, coefficients and enthalpy change labels; [1 mark]

[Total: 3 marks]

- Since these are enthalpies of formation the arrows must point from the elements in their standard states to the compounds
- Pay attention to multiples of the enthalpies changes or you may lose the mark
- Make sure you have the correct state symbols that match the standard state of the element as the cycle uses enthalpies of formation which by definition must start from the elements in their standard states

1b

b) Calculating the standard enthalpy change for the hydrolysis of urea:

- ΔH_r = 607.5 - 576.5 = 31 kJ mol^{-1}; [1 mark]

[Total: 1 mark]

- **Remember:** As you go around the cycle using Hess's Law, add the terms if you follow the arrow and subtract the terms if you go against the arrow
  - Using brackets around the terms helps to avoid careless calculation errors
- The workings are: ΔH_r = -(-287.0) - (-320.5) + (-414.5) + (-81.0 x 2)

1c

c) Calculating the enthalpy of combustion of urea:

$\Delta H_c = \sum \Delta H_f \, products - \sum \Delta H_f \, reactants$

- ΔH_c = (-829) + (-1148) - (-641) = -1336 kJ mol^{-1}; [1 mark]

[Total: 1 mark]

- The trick here is realising the method to solve this question is the same as the previous question, i.e. using the enthalpies of formation to solve the problem
- The relevant enthalpies of formation are:
  - Oxygen and nitrogen are both 0 kJ mol^{-1} as they are elements
  - ΔH_f for $CO(NH_2)_2$ is -320.5 kJ mol-1
    - **BUT** there are 2 moles used in the balanced equation, so 2 x -320.5 = -641 kJ mol^{-1}
  - ΔH_f for CO_2 is -414.5 kJ mol-1
    - **BUT** there are 2 moles used in the balanced equation, so 2 x -414.5 = -829 kJ mol^{-1}
  - ΔH_f for H2O is -287 kJ mol-1
    - **BUT** there are 4 moles used in the balanced equation, so 4 x -287 = -1148 kJ mol^{-1}

- Instead of using a cycle, you can use the relationship:
  - $\Delta Hr = \sum \Delta H_f \, products - \sum \Delta H_f \, reactants$

    OR

    $\Delta H_c = \sum \Delta H_f \, products - \sum \Delta H_f \, reactants$

  - Since the reaction is a combustion reaction
- ΔHr = (-414.5 x 2) + (-287.0 x 4) - (-320.5 x 2)
- ΔHr = (-829) - (-1148) - (-641)
- ΔHr = -1336 kJ mol^{-1}

2a

a) The energy change, in J, when 12.04 g of ammonium chloride is dissolved in 125.0 cm^3 of water:

- $n(NH_4Cl) = \dfrac{12.04 \text{ g}}{53.5 \text{ g mol}^{-1}} = 0.225$ (mol); [1]
- q = 0.225 mol x 15 100 J mol^{-1} = 3 397.5 (J); [1]

[Total: 2 marks]

- You need to work 'backwards' from the ΔH_{sol} given in the question to work out the energy change when 12.04 g or 0.225 mol of ammonium chloride is used
- This will give you a value in kJ mol^{-1} for q, but the question was asked for q in J

2b

b) To determine the value in temperature, in °C, when the lithium chloride is dissolved:

- $\Delta T = \dfrac{q}{mc} = \dfrac{3397.5 \text{ J mol}^{-1}}{125.0 \text{ cm}^3 \text{ x } 4.18 \text{ J K}^{-1} \text{ mol}^{-1}} = 6.5$ (°C); [1]

[Total 1 mark]

- Rearrange the $q = mc\Delta T$ equation to give you the temperature change

2c

c) To determine the final temperature, in °C, of the solution during the reaction:

- 19.5 °C - 7.1 °C = 13.0 °C; [1]

[Total: 1 mark]

- The positive sign on the enthalpy of solution value tells you that the reaction is

endothermic so the temperature will decrease
- In other words, you must subtract the enthalpy change from the initial temperature

3a

a) To determine the enthalpy of hydrogenation of propene:

- Hydrogenation equation: $CH_3CH=CH_2 + H_2 \rightarrow C_3H_8$; [1 mark]

Calculation option 1:

- ΔH_r = (C-C) + (C=C) + 6(C-H) +(H-H) -2(C-C) -8(C-H); [1 mark]
- ΔH_r = (347) + (619) + 6(413) + 435 - 2(347) -8(413) = -119 kJ mol^{-1}; [1 mark]

OR

Calculation option 2:

- ΔH_r = (C=C) + (H-H) - (C-C) - 2(C-H); [1 mark]
- ΔH_r = (619) + (435) - (347) -2(413) = -119 kJ mol^{-1}; [1 mark]

[Total: 3 marks]

- For this question, you need to know the structure of propene and that hydrogenation involves a reaction with hydrogen
- Always look for shortcuts to save you time
  - E.g. if the same bond is broken and formed, then there is no need to include it in the calculation

3b

b) Suggest, with a reason, whether the polymerisation of propene is exothermic or endothermic:

- One C=C bond is broken and two C-C bonds are formed;
  OR
  One C=C bond is is weaker / 619 $kJmol^{-1}$ than two C-C bonds / 2 x 347 = 694 $kJmol^{-1}$;
  [1 mark]
- There is a net output of energy / the reaction is exothermic; [1 mark]

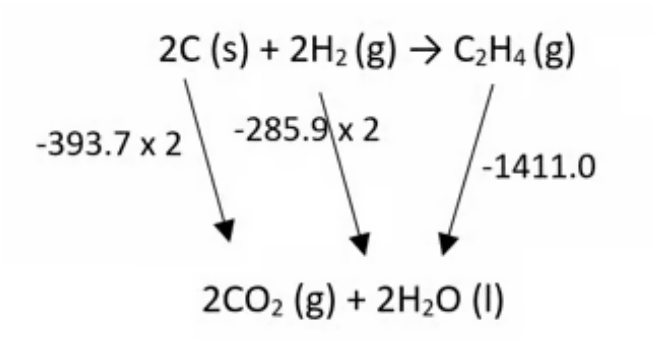
[Total: 2 marks]

- The polymerisation of propene would be:
  - $n CH_3CH{=}CH_2 \rightarrow n$ -[-CH(CH_3)CH_2-]-
- For *n* molecules of propene:
  - *n* C=C bonds are broken
  - 2*n* C-C bonds are formed
- The net energy change is the equivalent of breaking one C=C bond and forming two C-C bonds, which is exothermic

3c

c) To calculate the standard enthalpy change of formation of ethene:

- Correctly labelled cycle



OR

ΔH^{θ}_f = 2(–393.7) + 2(–285.9) – (–1411.0); [1 mark]

- ΔH^{θ}_f = + 51.8 kJ mol^{-1}; [1 mark]

[Total: 2 marks]

- The arrows must point downwards in the cycle and have the correct coefficients
- The wrong sign for -1411 would lose the first mark